MUSTANG III PILOT'S NOTES

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The carriage of wing bombs or drop tanks with full fuselage tank reduces this instability; hence should it be necessary to jettison wing loads while still carrying fuel in the fuselage tank, the pilot should expect a sudden change of trim and increase of instability.

(ii) *Change of trim*

Undercarriage down	Nose down
Flaps down	Nose down
Radiator shutter open ..	Nose down

The directional trim changes as speed and power are varied.

(iii) The elevator and rudder trimming tab controls are sensitive and must be used with care.

(iv) *Flying at low altitude in rain or bad visibility.*—Lower flaps to 20°, reduce speed to 150 m.p.h. I.A.S., and increase r.p.m. to 2,700. Check radiator and oil temperatures frequently.

43. Stalling

(i) The stalling speeds, engine off, are:

Flaps and undercarriage up ..	90 m.p.h. I.A.S.
Flaps 20° down, undercarriage down	85 ,, ,,
Flaps fully down, undercarriage down	75 ,, ,,

(ii) When no fuel is carried in the fuselage tank, slight tail buffeting occurs about 3 to 4 m.p.h. above the stall at which the right wing drops gently. With fuselage tank full or half full, there is no buffeting to give warning of the approaching stall, but a series of stick reversals occurs just above stalling speed, at the stall the right wing drops sharply, and unless immediate recovery action is taken, a spin may develop. The control column must be pushed firmly forward for recovery.

(iii) The aircraft sinks rapidly as stalling speed is approached.

(iv) If the control column is held back at the stall, a wing will drop very rapidly and the aircraft will become inverted.

(v) *High speed stall.*—If the aircraft is stalled in a steep turn either wing will drop very rapidly. The stall is preceded by pronounced juddering. Recovery is immediate if the pressure on the control column is relaxed.

54. **Position error correction.**

Corrections for position error are as follows:

From To	120 150	150 190	190 240	240 300	300 350	350 400	m.p.h. I.A.S. ,, ,,
Add Subtract	6	3	0	3	6	9	m.p.h. ,,

55. **Maximum performance**

(i) Speeds for maximum rate of climb:

160 m.p.h. I.A.S. from SL to 30,000 feet
155 m.p.h. I.A.S. from 30,000 to 35,000 feet
150 m.p.h. I.A.S. from 35,000 to 40,000 feet
145 m.p.h. I.A.S. above 40,000 feet

(ii) *Combat*

Set supercharger gear change to AUTO, r.p.m. to 3,000, and throttle lever fully forward.

56. **Maximum range**

(i) Estimated recommended speed 185 m.p.h. I.A.S.

(ii) To obtain any desired I.A.S., fly at 1,800 r.p.m. with supercharger switch LOW and obtain the I.A.S. by adjusting boost, but do not exceed 46 inches. If the desired speed cannot be obtained at 46 inches or full throttle, increase r.p.m. as necessary up to 2,700. At high altitudes, if the I.A.S. cannot be obtained at 2,700 r.p.m., set supercharger switch to AUTO.

When flying at low r.p.m. it will probably be necessary to clear the engine every 15 minutes by increasing r.p.m. to 2,700 for two or three minutes.

57. **Fuel consumption and capacity**

(i) Consumptions in the rich mixture range:

Boost	R.p.m.	Gallons/hour Imperial	*U.S.*
67	3,000	137	*165*
61	3,000	112	*135*